

Onyx — Developer Guide

Onyx · multi-process OS on Raspberry Pi 4

This guide explains how to **build** Onyx, how to **write an application** or a **/bin tool**, how to **extend the kapi ABI**, and the **conventions** + **pitfalls** to be aware of. For the details of how things work internally, see [Kernel internals](#).

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1. Prerequisites and toolchain

- **AArch64 bare-metal toolchain:** aarch64-none-elf- (GCC), used under **WSL** on a Windows development machine.
- **Circle**, pulled in as a **git submodule** at circle/ (our fork [stephaneweg/circle](#), branch `onyx`).
- `make`, `cp`, `mkdir` (standard Unix tools, via WSL).
- A **FAT32 SD card** and a **Raspberry Pi 4** to run on (there is no QEMU `raspi4b` in the reference dev environment; bring-up is done directly on hardware).

Important — which Circle? `circle/` is a **git submodule** → our fork [stephaneweg/circle](#) (branch `onyx`), **not** any other clone. Patch Circle *there* (commit on `onyx`), rebuild the affected library, then record the new commit in the superproject (`git add circle && git commit`). The patches are listed in [Circle Changes](#).

2. Building Circle (once)

```
# Circle is a git submodule (the fork stephaneweg/circle, branch onyx). Fetch it
# (or clone Onyx with --recurse-submodules in the first place):
git submodule update --init --recursive
```

```
# Configure for RPi 4 / AArch64. DEPTH=32 is REQUIRED: our GImage renders 32-bit
# pixels (Circle defaults to 16).
cd circle && ./configure -r 4 -p aarch64-none-elf- -d DEPTH=32 -f
```

```
# Build the libraries used by the kernel:
(cd lib && make -j4) \
&& (cd lib/sched && make -j4) \
&& (cd lib/fs && make -j4 && cd fat && make -j4) \
&& (cd lib/usb && make -j4) \
&& (cd lib/input && make -j4) \
&& (cd addon/SDCard && make -j4) \
&& (cd addon/fatfs && make -j4)
cd ..
```

If you change DEPTH later, run `make clean` in `circle/lib` before rebuilding (the `.o` files do not depend on `Config.mk`).

The libraries linked by the kernel (cf. `kernel/Makefile`): `libsched.a`, `libfatfs.a`, `libsdcard.a`, `libfs.a`, `libusb.a`, `libinput.a`, `libcircle.a`.

The Onyx-specific patches carried by this fork (branch `onyx`, on upstream tag `Step51`) are documented in [Circle Changes](#).

3. Building the kernel and applications

From `kernel/`:

```
cd kernel
make          # -> kernel8-rpi4.img, THEN builds the apps (../user)
```

The default target (`all`):

1. compiles our sources + links against the Circle libs → `kernel8-rpi4.img`;
2. triggers `make -C ../user`, which builds **all the apps** (`user/*.elf`) and the **/bin tools** (`user/bin/*.elf`).

Kernel → apps order. Since the apps go through the **fixed-address ABI table** and are **not** linked against the kernel's addresses, they no longer depend on the kernel build: a *kernel-only* change (that respects the ABI) does **not** require recompiling the apps. `make` rebuilds them anyway for convenience.

Details of the kernel wiring (which Circle files we replace, the order of the `-I compat` `-I include` include path before `circle/include`): see [Kernel internals §1](#) and `kernel/Makefile`.

4. Deploying to the SD card

```
cd kernel
make stage      # copie l'image + chaque app + chaque outil vers ../sdcard/
```

`make stage`:

- copies `kernel8-rpi4.img` → `sdcard/`;
- for each `../user/<name>.elf`: creates `sdcard/apps/<name>.app/` and copies the ELF into it as `main.elf` (the "Onyx" layout);
- copies each `../user/bin/<tool>.elf` → `sdcard/bin/<tool>.elf`.

Then copy **all** of the contents of `sdcard/` onto a **FAT32** card and boot the Pi 4. See the [user guide](#) for details about the card and the configuration files.

Keyboard layouts live as binary files in `sdcard/etc/keymaps/<NAME>.kmap` (the `keyb` tool — and the theme editor's dropdown, which lists whatever `.kmap` files are actually present — load one and send it to the kernel via `kapi_set_keymap_data`, v27). **The kernel compiles in no keyboard map at all:** at boot the keyboard table is empty and stays so until `keyb` loads a layout from the card (see [Circle Changes §1](#)), so a layout is *only* ever a file — adding one needs no kernel rebuild. Regenerate the `.kmap` files with `python tools/keymaps/genkeymaps.py`; it compiles each `<NAME>` from a `keymap_<name>.h` table in `tools/keymaps/maps/` (the 8 layout sources, incl. `BE` = Belgian azerty, all tracked in the Onyx repo — not in the Circle tree). The `.kmap` format is `"OKM1"` + `u16` rows/cols + the `u16[128][5]` table (see the script header).

5. The application model

A Onyx application is a **single .c file** compiled into a **freestanding ELF** and run in **EL1** in its own page table.

Runtime (`user/crt0.S`):

```
_start:
    stp x29, x30, [sp, #-16]!    /* the stack is already set up by the kernel */
    bl  main                    /* call main() */
    ldp x29, x30, [sp], #16
    ret                        /* return -> the kernel terminates the task */
```

No `argv` is passed via the stack: `main()` takes no arguments. An app retrieves its argument line via `kapi_get_args(buf, size)`, and exits early with `kapi_exit(status)`.

Linking (`user/user.ld`): everything is linked at `USER_VA_BASE = 8 GB` (`. = 0x200000000`;). This is **not PIE**. The `RX` (`.text/.rodata`) and `RW` (`.data/.bss`) sections are aligned on **64 KB** (`-z max-page-size=0x10000`) so that the loader maps them onto distinct pages. **No kernel symbol** is resolved at link time: everything goes through the ABI table.

Compilation flags (`user/Makefile`):

```
-ffreestanding -nostdlib -fno-pic -fno-pie -mgeneral-regs-only \
-O2 -Wall -Wextra -fno-stack-protector -I../kernel/include
```

- `-ffreestanding -nostdlib`: **no libc**. No `printf`, `malloc`, `string.h`... use `applib.h` (§8) and static/local buffers.
- `-mgeneral-regs-only`: **no FP/SIMD** — the kernel trap frame does not save the floating-point registers. Do **integer / fixed-point arithmetic** (see `tinycalc.c`, `mandelbrot.c`).
- `-I../kernel/include`: to include `<kern/kapi_abi.h>` (the shared ABI structure).

To add an app, create `user/<name>.c` and **add `<name>.elf` to the `PROGS` variable** of `user/Makefile` (and `<tool>.elf` to the `PROGS` of `user/bin/Makefile` for a tool).

6. Writing a graphical application

Minimal skeleton (window with kernel-managed widgets):

```
#include "kapi.h"

static void on_ok (unsigned long sender, int ev, long value)
{
    (void) sender; (void) ev; (void) value;
```

```

    kapi_widget_set_text (g_label, "cliqué !");
}

static unsigned long g_label;

int main (void)
{
    /* Le canvas est mappé à 12 Go ; fb[y*w + x] = 0x00RRGGBB. */
    unsigned *fb = kapi_create_window (300, 200, "exemple");

    g_label = kapi_add_label (10, 10, 200, 16, "prêt");
    (void) kapi_add_button (10, 40, 80, 28, "OK", on_ok);

    kapi_wait_for_exit (); /* pompe Les événements à ~60 fps jusqu'à fermeture */
    return 0;
}

```

Key points:

- **kapi_create_window(w, h, title)** returns a pointer to the **canvas** (pixel buffer of 0x00RRGGBB, width w). The app draws directly into it (no per-pixel call). The variant **kapi_create_window_ex(x, y, w, h, title, flags)** is for explicit placement and **WIN_FLAG_BORDERLESS**.
- **Kernel widgets:** **kapi_add_button/label/checkbox/textbox/progress/slider/textarea/scrollbar_v/scrollbar_h/icon(...)** return an **unsigned long** handle. Manipulate them with **kapi_widget_set_text/get_text, get_checked, get/set_value, set_rect** (move/hide by setting **w=h=0**), **set_icon**.
- **Event loop:** either **kapi_wait_for_exit()** (blocking, simple), or your own loop **while (!kapi_should_exit()) { ...; kapi_pump_events(); kapi_present(); kapi_msleep(16); }** when you animate the canvas yourself.
- **App-drawn UI:** to draw text over your canvas, use **kapi_draw_text(x, y, s, color) + kapi_font_width/height()**. Capture the keyboard with **kapi_set_key_handler(fn)** (**GUI_EVENT_KEY** events, **KEY_*/ASCII** values) and the "outside-widget" clicks with **kapi_set_click_handler(fn)** (**GUI_EVENT_CANVAS_CLICK / ..._MOTION**, coordinates encoded in value). Cursor position relative to the window: **kapi_cursor_pos(&x, &y)**.
- **Cooperative:** your app **must yield** regularly (**present/msleep/ pump_events/wait_for_exit**), otherwise it freezes the whole system.

See the demos **demoD.c** (widget gallery), **demoE.c** (textarea + scrollview), and the apps **tinypad.c**, **paint.c**, **mandelbrot.c** for complete examples.

7. Writing a /bin tool

A **/bin** tool follows the **same EL1 app model** but reads **stdin**, writes **stdout**, and exits (no window). It is composable via the terminal's pipes.

```

#include "kapi.h"
#include "applib.h" /* ax_puts, ax_putln, ax_strlen, ax_itoa, ... */

int main (void)
{
    char args[128];
    kapi_get_args (args, sizeof args); /* La ligne d'arguments (chaîne) */
}

```

```

/* Lire stdin et Le réécrire en majuscules, par exemple : */
char buf[256];
int n;
while ((n = kapi_stdin_read (buf, sizeof buf)) > 0)
{
    for (int i = 0; i < n; i++)
        if (buf[i] >= 'a' && buf[i] <= 'z') buf[i] -= 32;
    kapi_stdout_write (buf, (unsigned) n);
}
return 0; /* code de sortie récupérable via kapi_wait dans l'appelant */
}

```

- **kapi_get_args(buf, size)**: the entire argument line as a **single string** separated by spaces (there is no argv array; parse the first word yourself, etc.).
- **kapi_stdin_read(buf, n)**: reads the task's stdin (0 = EOF). **kapi_stdout_write**: writes stdout.
- To read a file passed as an argument: **kapi_open/read/close**. To list a directory: **kapi_opendir/readdir/closedir**.

The existing tools to study: **ls, cat, grep, wc, echo, page, rm, mkdir, touch, cp, mv, ps, kill, run, keyb** (in user/bin/).

8. The **applib.h** library

user/applib.h is **header-only** (no libc). It provides:

- **Strings**: **ax_strlen**, **ax_streq**, **ax_strcat(dst, cap, &pos, src)** (concat without overflow), **ax_app_path(dst, cap, name, suffix)** (builds SD:apps/<name><suffix>), **ax_itoa(v, buf)**, **ax_fmt2(d, v)** (2-digit decimal).
- **Console**: **ax_puts(s)**, **ax_putln(s)** (to stdout).
- A minimal **.ini reader**: **app_ini_load("config.ini")** (from the app's folder via **kapi_app_dir**) or **app_ini_load_path("SD:skins/theme.txt")**; then **app_ini_get(section, key, default)** and **app_ini_get_int(...)**. Sections [xxx], lines key=value, comments ;/##.
- **App-drawn widgets** (not kernel widgets): **ax_dropdown** (drop-down list) and **ax_colorpick** (palette-based color picker), with ***_draw(...)** and ***_click(...)**. The app draws them in its canvas and routes the clicks via its **set_click_handler**. Used by **theme.c** and **mandelbrot.c**.

There is also **user/httpc.h** — a header-only **HTTP/1.0 client** over the v21 TCP socket calls. It does **no allocation** (the caller passes the response buffer, so all memory stays in the app's address space): **http_get(url, buf, cap, &resp)**, **http_post(...)**, or the general **http_request(method, url, headers, body, len, buf, cap, &resp)**. Plain HTTP only (no TLS). Used by **/bin/wget**. This is the recommended pattern for application-level protocols: build them in a user library on top of the kernel's transport kapis, rather than adding them to the ABI.

And **user/uikit.h** — a **retained-mode widget toolkit** drawn entirely in the app's canvas, driven by the kernel's **pointer stream** (ABI v22: **set_pointer_handler** → **GUI_EVENT_PTR_MOVE/DOWN/UP/ENTER/LEAVE** with client coords). Same memory model as the rest: widgets live in a caller-provided **fixed pool** (**ui_widget pool[N]** in the app's .bss, freed automatically on exit — no user **malloc**, no kernel object behind a widget). **ui_init**, **ui_button/ui_label/ui_checkbox/ui_textbox**, **ui_on_event** (fed from the app's pointer + key handlers), **ui_draw**. The **textbox** is a single-line editor: caret, Backspace/Delete, arrows, Home/End, Tab to move focus, an optional password mask (**ui_set_password**), read with **ui_get_text**. **tinycalc** (buttons) and **wpaconf** (a form of textboxes) use the toolkit. This is the forward

path for widgets: new ones are added here, in userland, with no kernel/ABI change. The older **kernel-drawn widgets** (`add_button...`, §earlier) still work and coexist; apps choose one model per window.

Dynamic memory + C++ apps

Apps have **no malloc by default** — they use static buffers + the stack (both in the app's address space). For dynamic memory, include `user/umm.h`: a small user allocator (size-class free lists + a `kapi_sbrk` arena). `umm_malloc` / `umm_free` / `umm_calloc` / `umm_realloc`. The heap lives at `USER_HEAP_BASE` (10 GB); its pages are owned by the address space, so they are **freed automatically when the app exits** and show up in the app's page count (`ps` / `memmon`). `kapi_sbrk` is the underlying primitive (rarely called directly). `/bin/heaptest` exercises it.

C++ apps are supported (freestanding subset — no exceptions, no RTTI, no STL):

- Name the source `*.cpp`; the user Makefile builds it with `g++ (-fno-exceptions -fno-rtti -fno-threadsafe-statics -fno-use-cxa-atexit)`.
- Include `user/onyxpp.hpp`: it defines operator `new/delete` (on `umm`) and the runtime stubs (`__cxa_pure_virtual`, `__dso_handle`, `__cxa_atexit/atexit` no-ops). Global constructors run via `crt0.S` (the `.init_array` walk); static destructors are **not** run (the app exits and its address space is reclaimed).
- Classes, inheritance and virtual methods (vtables) work; `new/delete` go through the user heap. No `std::string/std::vector` — write small containers on `umm` as needed. See `user/cppdemo.cpp` for a working example.

9. Packaging an app: .app, icons, config.ini

Layout of an application on the card (produced by `make stage`):

```
SD:apps/<nom>.app/
main.elf      l'ELF de l'app (obligatoire)
icon.bmp      icône 40×40, BMP 24 bpp ; le magenta 0xFF00FF est transparent (optionnel)
app.txt       métadonnées (display name + catégorie) lues par le launcher (optionnel)
config.ini    configuration de l'app, lue via app_ini_load() (optionnel)
```

The **app name** is the base name of the `.app` folder (without the suffix). That is what you put in `/etc/autostart` / `/etc/quicklaunch.txt` and what `kapi_list_apps` returns.

app.txt — friendly metadata for launchers (key = value, no section, read with `app_ini_load_path` / `app_ini_get(0, ...)`). Every app under `SD:apps/` ships one:

```
name      = Text Editor           ; display name shown under the icon
category  = Productivity          ; Games, Graphics, Productivity, Internet, System, Demos, Shell
```

The current app-drawer (`applist`) still labels icons by folder name; `app.txt` is the groundwork for a **category-grouping launcher** (groups icons by `category`, shows `name`). The two shell components `panel/applist` carry `category = Shell` so a launcher can exclude them.

Icons — `tools/gen_assets.py` procedurally generates the 40×40 BMPs (BGR bottom-up, 4-byte padding) for all the apps (a document for `tinypad`, a calculator for `tinycalc`, a glider for `life`, etc.) and the "9 squares" glyph of the `apps` button in the panel. `tools/preview_icons.py` produces a PNG preview montage. Workflow: run `gen_assets.py` (writes the `icon.bmp` files into `sdcard/apps/<name>.app/`), then `preview_icons.py` to check visually.

`config.ini` — example read by `inidemo` / `voronoy` / `panel`:

```
; commentaire
greeting = bonjour
[app]
name = Mon App
version = 1
[display]
barwidth = 40
```

Screenshots and documentation

- **Screenshots (simulated but faithful):** `tools/screenshot/render.py` loads the **real** skins (`SD:skins/wings/button/closebgs.bmp`), the **real** font `circle/lib/font8x16.cpp` and the icons `SD:apps/<name>.app/icon.bmp`, and reproduces the drawing routine of each app — so the output matches the real OS. Run `python tools/screenshot/render.py` → writes the PNGs into `screenshots/` (`desktop.png` = overview, plus one screenshot per app, window only on a transparent background). **To add an app:** write `app_<name>()` that reproduces the `redraw()` of `user/<name>.c` (same colors/coords), add `("<name>", app_<name>)` to the APPS list, and re-run.
- **Word/PDF exports:** `docs/build_docs.py` converts each `.md` in `docs/` into `.docx` (via `pandoc`) then into `.pdf` (via `Word/docx2pdf`), in `docs/exports/`. Run `python docs/build_docs.py` after any modification to the `.md` files. Prerequisites (once): `pip install python-docx docx2pdf pypandoc_binary`.
- **Reminder of the project rule** (see `CLAUDE.md`): any modification of a `kapi` function or of an app **updates the docs in the same go**, regenerates the affected screenshot if the visual changes, then regenerates the exports.

10. Extending the `kapi` ABI

The ABI is **append-only**. To add a kernel function callable by apps, there are **5 points** to touch (all in the same direction, at the end):

1. `kernel/include/kern/kapi_abi.h` — add the function pointer **at the end** of `struct TKApiTable` (never in the middle, never reorder), with a version comment, and **increment** `KAPI_ABI_VERSION`.

```
// --- v22 additions --- (next free version; v21 added the TCP sockets)
int (*ma_fonction) (int arg);
```

2. `kernel/sys/kapi.cpp` — implement `extern "C" int kapi_ma_fonction(int arg)`. It runs in the app's context (use `CurrentAS()` for the current address space if needed).
3. `kernel/sys/kapitable.cpp` — declare the `extern "C"` prototype and **assign the pointer** in `KApiTableInit()`: `t->ma_fonction = kapi_ma_fonction;`
4. `user/kapi.h` — add the inline wrapper:

```
static inline int kapi_ma_fonction (int arg) { return KT->ma_fonction (arg); }
```

5. Rebuild (`make`). The apps that want the new function use it; the old ones keep working (they ignore the new field).

Golden rule: never change the signature or the order of an existing field. If some semantics must change, add a **new** entry. An app can query `((const struct TKApiTable *)KAPI_TABLE_VA)->version` to find out what is available.

If you add a new **GUI event** or a **window flag**, keep the values synchronized between `kernel/gui/window.h` and the `#defines` in `user/kapi.h` (commented "must match").

11. Coding conventions

- **Kernel (C++)**: Circle style. `Cxxx` classes, `m_Xxx` members, CamelCase methods, `boolean/TRUE/FALSE` and Circle's `u8/u16/u32/u64` types. No exceptions or RTTI. `new/delete` go through Circle's heap.
- **Userland (C)**: freestanding C. `ax_` prefix for the `applib.h` helpers. Globals `g_xxx`. Bounded static buffers (no dynamic allocation on the app side in general).
- **kapi**: `extern "C"` functions named `kapi_xxx` on the kernel side; inline wrappers `kapi_xxx` on the app side.
- Respect the **comment density** and the **idiom** of the file you are modifying.
- **Git**: commit into the **Onyx repo** explicitly — the current working directory (cwd) drifts; a bare `git` may land in the wrong repo. (Note: `circle/` is not committed in this repo.)

12. Debugging on hardware

Bring-up is done **directly on the Pi 4** (no QEMU raspi4b). Tools:

- **On-screen exception dump**: an EL1 synchronous fault (or an EL0 fault) paints a panic + register dump on the HDMI framebuffer (`PanicToScreen` + Circle's handler). Note the `ELR` (faulting PC).
- **addr2line**: `aarch64-none-elf-addr2line -e kernel8-rpi4.elf <ELR>` to locate the faulting line (keep the unstripped `.elf` next to the `.img`).
- **Serial console**: `config.txt` must have `enable_uart=1` (PL011 clock). The boot log goes **also** to the HDMI screen (`CScreenDevice`) so it is readable without a serial cable.
- **Post-mortem console**: on an app's exit, the compositor clears and the logger is shown on the framebuffer (see [Kernel internals §11](#)).

13. Known pitfalls

- **Do not save FP/SIMD**. `-mgeneral-regs-only` is mandatory on the app side; use fixed-point. An accidental floating-point use would corrupt the state on a context switch.
- **L3 tables shared with the kernel**. On the kernel side, never free an L3 table from the user area without checking that it is not shared with the kernel's L2 (cf. [Kernel internals §4](#)). Otherwise: global corruption.
- **Do not free the ABI table page**. It is global to the kernel; the destruction of an address space already skips it.
- **DEPTH=32 for Circle**. `GImage` renders 32-bit; forgetting `-d DEPTH=32` (or changing `DEPTH` without `make clean` in `circle/lib`) gives wrong colors/breakage.
- **Cooperative**. Any app loop without `present/msleep/yield` freezes the system (no preemption).
- **Circle LF renormalization**. On Windows, Circle is checked out in CRLF; renormalize once (cf. §2) otherwise the build breaks.
- **The right Circle**. Patch `Zircon/circle`, not another clone.